When Big Firms Stumble, Startups Pay

Category Spillover and Startup Responses

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INTRODUCTION

Large-scale breaches of personal data have become a recurring feature of corporate life

147M

people whose personal information was exposed in the 2017 Equifax breach

$5B

FTC penalty on Facebook for privacy violations surrounding Cambridge Analytica (2019)

190M

people whose data was compromised in the 2024 Change Healthcare cyberattack

3,158

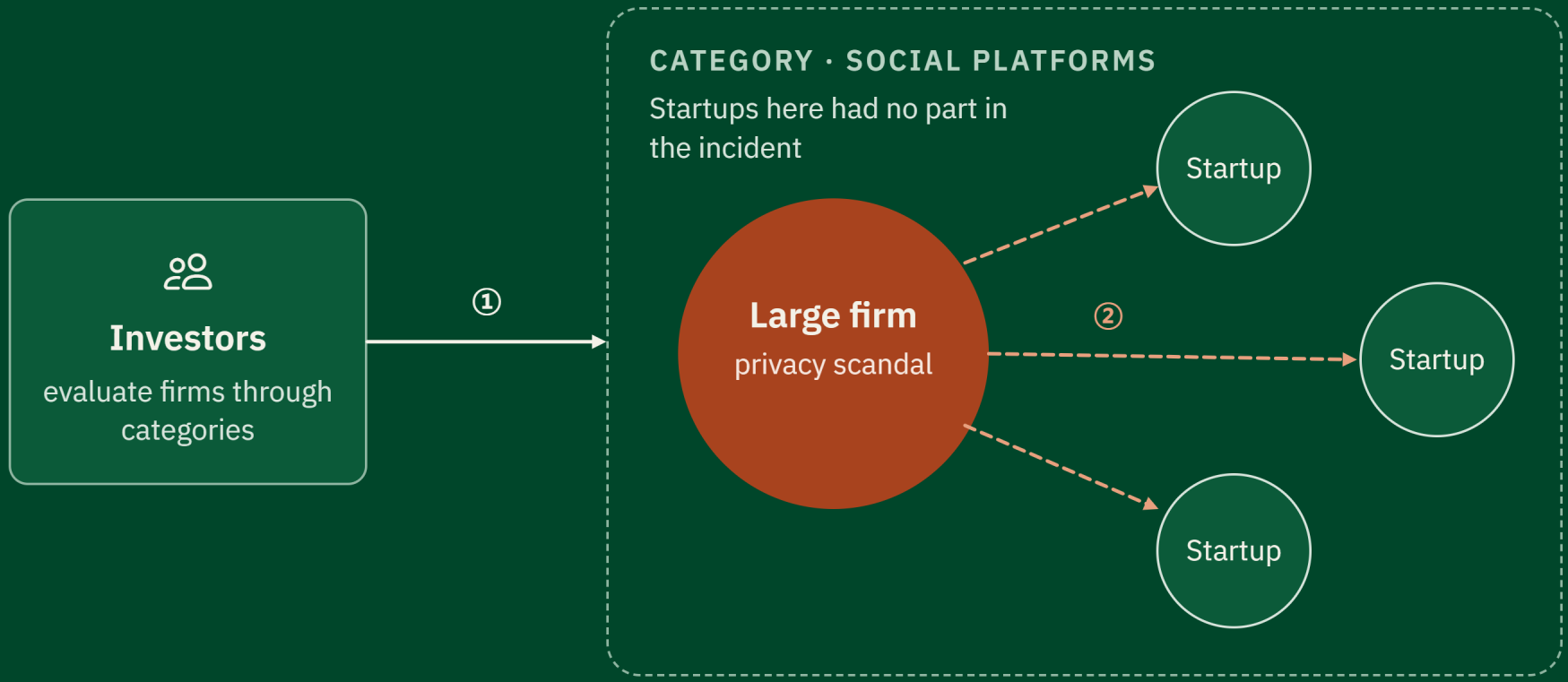
data compromises reported by U.S. organizations in 2024, with 1.3B+ victim notices

> When such incidents occur, public attention converges on the large firm responsible, and the subsequent debate centers on how that firm should be punished and regulated. *Yet the consequences of an incident are not confined to the firm that caused it.*

FTC (2019a, 2019b); Infosecurity Magazine (2025); Identity Theft Resource Center (2025)

CENTRAL ARGUMENT

A category is both a foundation of legitimacy for startups *and* a channel through which other firms' misconduct spreads.



① Foundation of legitimacy

Evaluators perceive firms through categories, groups of firms regarded as the same kind (Zuckerman, 1999).

② Channel of misconduct

What happens to one member of a category spills over to the others (Kostova & Zaheer, 1999).

PRIOR RESEARCH

The closer a firm is to the offender, the more of the spillover it bears?

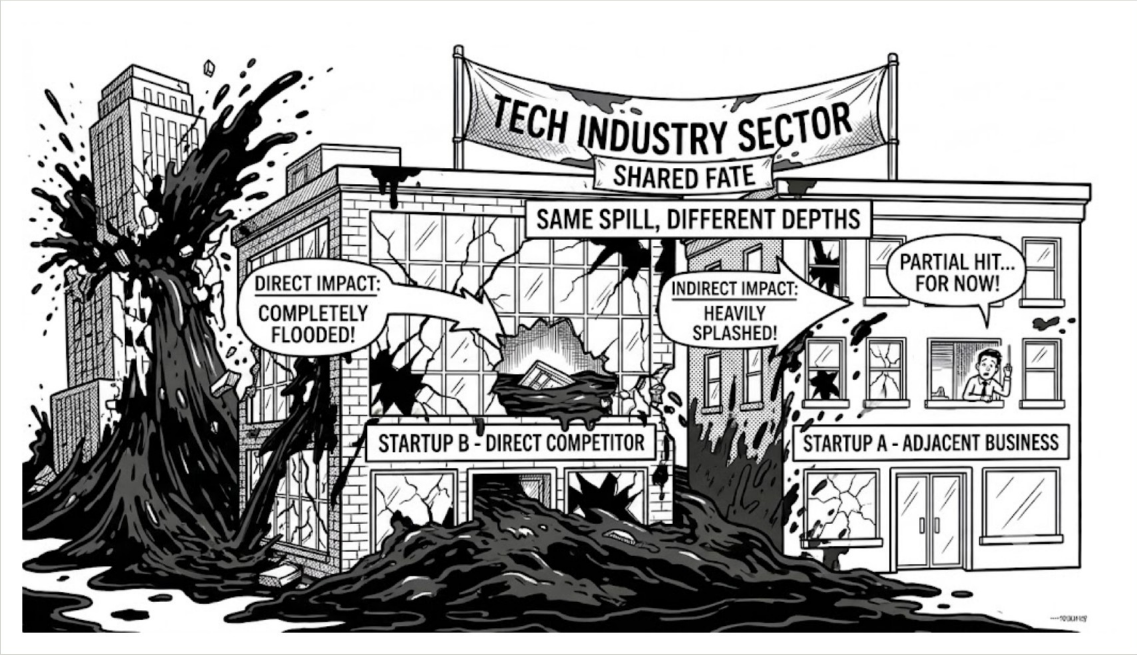


Illustration of the resemblance-based prediction

Prior research has defined closeness as **resemblance** to the offender in business or organizational form (Kostova & Zaheer, 1999; Jonsson et al., 2009) or as **holding the attribute implicated** in the incident (Diestre & Rajagopalan, 2014; Naumovska & Zajac, 2022).

> *Yet firms that hold the same attribute can differ in its depth.*

RESEARCH GAPS

Two open questions — both matter most for startups

QUESTION 1

What makes a blameless firm close to the offender?

Resemblance and depth need not go together, and if evaluators follow depth, the firms that resemble the offender most may not be the ones that suffer most.

QUESTION 2

Which firms in a category bear less of a spillover?

The known sources of protection are means available to established firms: industry self-regulation, endorsements from industry associations, and exit from the stigmatized business.

RESEARCH QUESTION

How does an incident at a large firm affect the fundraising of startups in the same category, and which startups within that category bear more or less of the penalty?

II

Theoretical Background and Hypotheses

2.1 Categories and Legitimacy Spillover

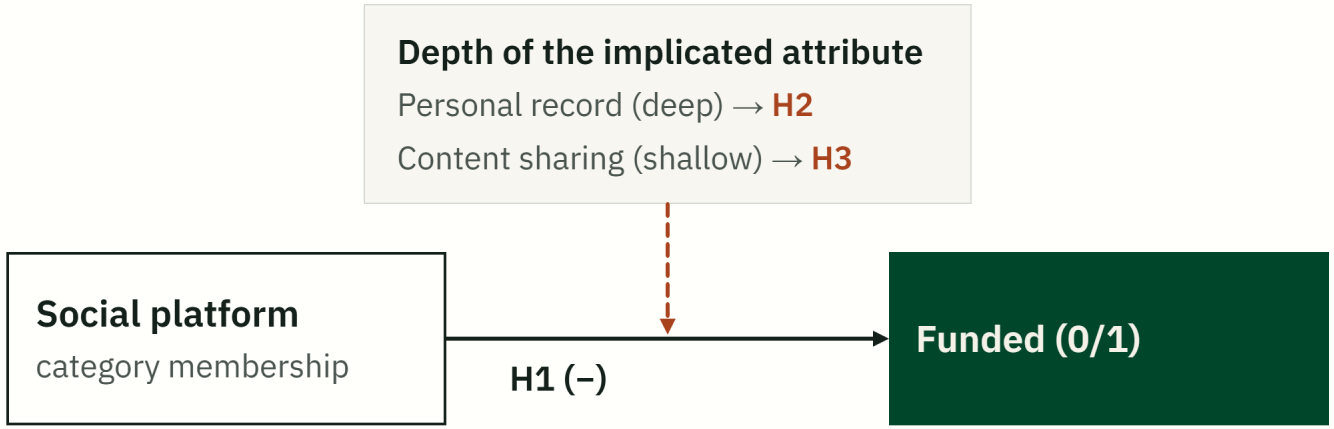
2.2 Stigma Spillover and Startups · H1

2.3 Depth of the Implicated Attribute · H2, H3

2.4 Distinguishability within a Category · H4

Research model

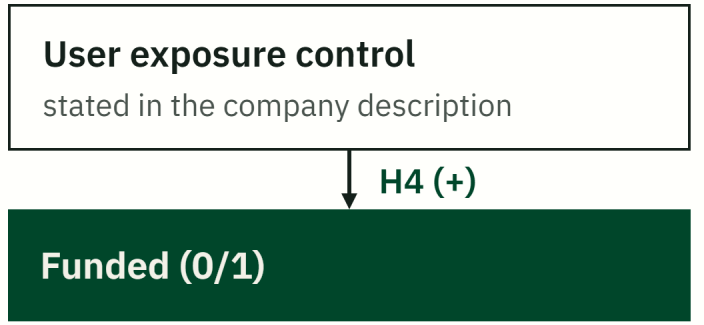
A · FULL SAMPLE · N = 51,738 · MODELS 2–4



Each interaction entered in a separate model; both service-type main effects in all models

B · WITHIN THE CATEGORY

Social platform startups only · N = 1,804 · Models 5–7



Estimated within social platforms

HYPOTHESIS 1

Startups in the same category as a large firm that caused a user-information misuse incident, that is, social platform startups, will be less likely to raise funding than other startups.

2.3 DEPTH OF THE IMPLICATED ATTRIBUTE

The degree to which the attribute, in the form a firm holds it, touches what the affected parties most want protected

Provided on the premise that it stays on the platform — **Already meant to be public**

Personal record services · deep

Users log information they would prefer others not see, such as their health conditions, emotions, and personal concerns, for their own use.

Content sharing services · shallow

Users post videos, music, photos, and other content intended from the outset to be seen by others; these services overlap most with Facebook's business.

HYPOTHESIS 2

The fundraising disadvantage of social platform startups will be greater for those in which user information runs deep.

HYPOTHESIS 3

The fundraising disadvantage of social platform startups will be smaller for those in which user information runs shallow.

Information sensitivity (Mothersbaugh et al., 2012; Milne et al., 2017) · Contextual integrity (Nissenbaum, 2004)

2.4 DISTINGUISHABILITY WITHIN A CATEGORY · H4

For a startup to be distinguishable, the information investors first encounter must show that the firm lacks the flaw

Spillover arises because evaluators cannot observe each firm's conduct and so cannot discriminate between blameless and flawed firms in a category (Barnett & King, 2008).

User exposure control refers to features that let users decide how much, and to whom, they are visible.

HYPOTHESIS 4

Among social platform startups, those that state user exposure control features will be more likely to raise funding than those that do not.

DESIGN TYPE (TABLE A3)	
Anonymous or pseudonymous participation	29
Invitation-based or private community	26
Visibility settings	11
Private Social Networking classification	63
At least one type (6.6%)	**121**

Excluded: ways in which platforms verify users or moderate content — there the platform, not the user, sets the scope of exposure

III Methods

3.1 Data
3.2 Research Context
3.3 Dependent Variable
3.4 Independent and Moderating Variables
3.5 Control Variables
3.6 Analysis

The scandal was caused by a single large platform, but it made visible the risks of the social platform category as a whole

December 2015

First reported (Davies, 2015); neither the firm nor regulators took official action

March 2018 · the event

Information on about 87 million users reported as passed to a political consulting firm; Facebook suspends CA; FTC announces investigation

May 2018

GDPR takes effect, two months after the scandal

July 2019

FTC imposes a $5 billion penalty on Facebook

Why this context

Most social platform startups had no direct connection to the incident. The context clearly separates parties to the incident from non-parties, allowing us to observe penalties that arise from **category membership alone**.

Social platforms

Online services in which users post information about themselves and form relationships or interact with other users through that information (boyd & Ellison, 2007; Kane et al., 2014).

Cadwalladr & Graham-Harrison (2018); Facebook (2018); FTC (2018, 2019b)

3.1 DATA

51,738 U.S. startups in Crunchbase founded before (2015–2017) or after (2019–2021) the scandal



About 44% of firms record only a founding year, with the date entered as January 1, so for firms founded in 2018 it is impossible to tell whether they were founded before or after the scandal: **61,618 − 9,880 = 51,738**. The full sample is used in robustness checks.

Crunchbase covers not only funded firms but also firms that did not raise funding, which reduces the problem of selecting the sample on the outcome (Dalle et al., 2017).

Funded	**15,883** (30.7%)
Social platform	**1,820** (3.5%)
… personal record	**71**
… content sharing	**101**
… user exposure control	**121** (6.6%)

3.3–3.5 VARIABLES

Measures from industries and descriptions

Variable	Measurement	Source
Funded (DV)	1 if total funding recorded in Crunchbase exceeds one dollar (firms without a funding record are recorded as 1)	Funding history
Social platform	Social Network, Social Media, Private Social Networking, or Dating	Industry class
Personal record	Wellness, Fitness, Mental Health, Health Care, Personal Health, Health Diagnostics, or Medical	Industry class
Content sharing	Media and Entertainment (no firm falls into both types)	Industry class
Exposure control	Expressions of anonymous or pseudonymous participation, private communities, or visibility settings (Table A3), or the Private Social Networking classification	Descriptions, cross-validated with websites
Controls	ln # patents; Silicon Valley; New England; female, white founder; nonprofit; Pledge 1%; # industry categories	Crunchbase

Validation (Table A6): self-recording terms 28.2% (personal record) vs. 4.0% (content sharing); shared-content terms 67.3% vs. 19.7%

3.6 ANALYSIS

Logistic regression with industry-group and founding-year fixed effects

> Because the dependent variable is binary, we used **logistic regression**. All models include the control variables, industry-group fixed effects, and founding-year fixed effects.

Industry-group FE

H1 tests not whether social platforms receive less funding than other industries, but whether, **within the same industry groups**, firms in the social platform category receive less funding than firms outside it (43 groups).

Model sequence

Model 1: controls only. Model 2: + social platform (H1). Models 3–4: each interaction entered one at a time (H2, H3). Models 5–7: H4, among social platform startups only.

Interpretation

In logit models, the size and significance of an interaction coefficient alone do not allow a straightforward interpretation (Ai & Norton, 2003; Hoetker, 2007); we report odds ratios and predicted probabilities.

Founding-year FE absorb year-to-year differences in the investment market and right-censoring

IV
Results

4.1 Main Results · H1–H4

4.2 Robustness Checks

4.3 Additional Analyses

RESULTS

Descriptive Statistics and Correlations

Table 1. Descriptive Statistics and Correlations

Variable	Mean	SD	Min	Max	1	2	3	4	5	6	7	8	9	10	11
1. Funded (0/1)	0.307	0.461	0	1											
2. Social platform (0/1)	0.035	0.184	0	1	−0.02										
3. Personal record (Wellness/Health)	0.131	0.337	0	1	0.10	−0.05									
4. Content sharing (Media & Ent.)	0.017	0.127	0	1	0.01	0.06	−0.05								
5. Nonprofit (0/1)	0.010	0.102	0	1	−0.04	−0.00	−0.01	−0.00							
6. Pledge 1%	0.003	0.056	0	1	0.02	−0.00	−0.01	−0.00	−0.01						
7. # Industry categories	3.187	1.887	0	23	0.15	0.12	0.04	0.07	−0.04	0.04					
8. ln # patents	0.082	0.406	0	10.5	0.18	−0.02	0.09	−0.01	−0.02	−0.00	0.05				
9. Female founder	0.110	0.313	0	1	0.11	0.03	0.07	0.01	0.01	0.04	0.10	0.02			
10. White founder	0.978	0.145	0	1	−0.09	−0.02	−0.01	−0.02	0.01	−0.01	−0.07	0.00	−0.13		
11. Silicon Valley	0.065	0.247	0	1	0.04	0.00	−0.02	−0.01	−0.01	−0.01	0.05	0.07	0.01	0.00	
12. New England	0.059	0.236	0	1	0.05	−0.02	0.05	−0.01	0.00	0.02	−0.01	0.05	0.01	0.02	−0.07

Note: N = 51,738 (main sample). The mean VIF is 1.10.

30.7%
raised funding

3.5%
social platforms

1.10
mean VIF

Table 1 · N = 51,738 (main sample)

4.1 MAIN RESULTS · H1 SUPPORTED

Social platform startups were less likely to raise funding than other startups in the same industry groups

−4.5 pp

lower probability of raising funding (average marginal effect, Model 2) — about 15% of the sample's mean funding rate (30.7%)

Social platform (Model 2)	**β = −0.248*** **
Odds of raising funding	**about 22% lower**
LPM, with industry-group FE	**−4.9 pp*** **
LPM, without industry-group FE	**−7.5 pp*** **

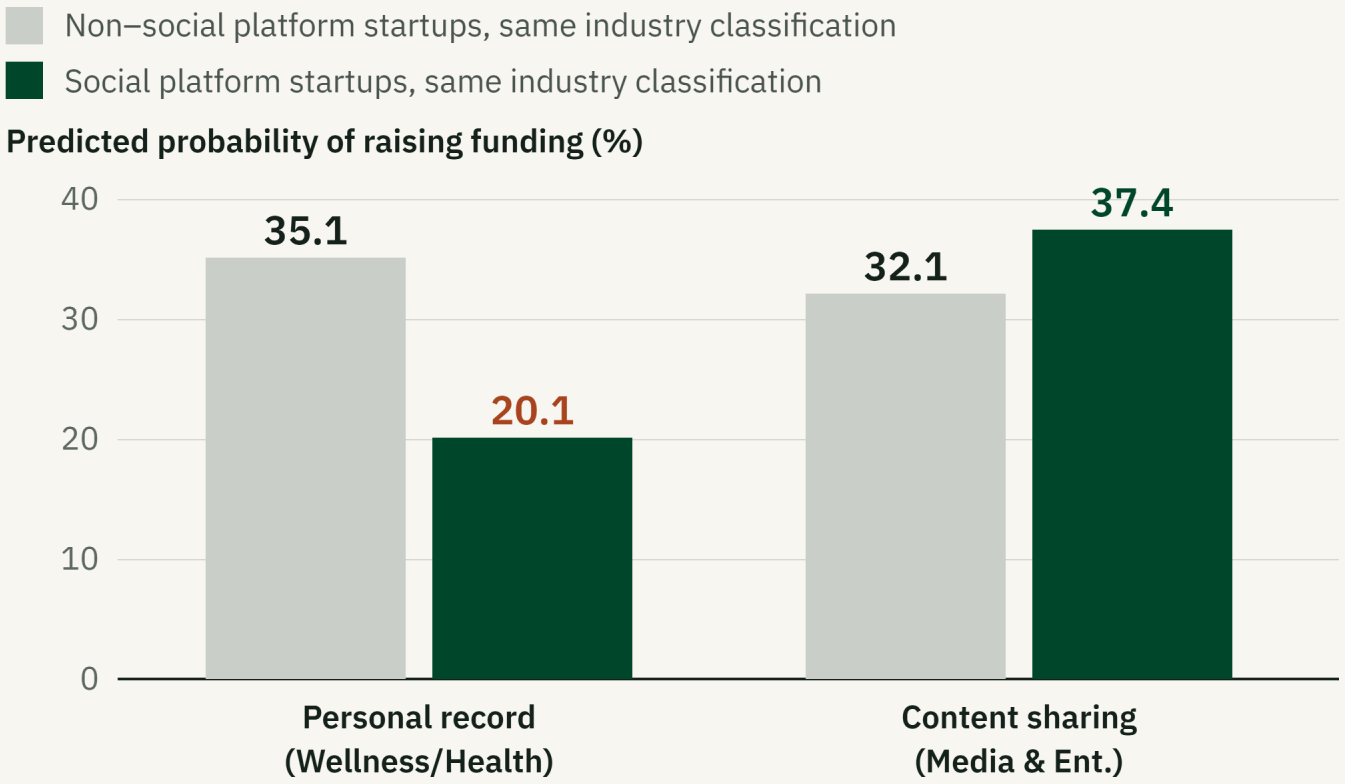
HOW TO READ

Compared with a startup in the same industry group and founding year that is not a social platform, a social platform startup's predicted probability of raising funding is 4.5 percentage points lower; its odds are exp(−0.248) ≈ 0.78 times as high. Without industry-group FE the gap appears larger (−7.5 pp), so part of the raw gap reflects industry differences that the main model removes.

Table 2, Model 2; Table A5 · pp = percentage points · *** p < 0.001

4.1 MAIN RESULTS · H2 AND H3 SUPPORTED

The penalty was greater for personal record services and statistically indistinguishable for content sharing services



H2 · PERSONAL RECORD

−15.0 pp

20.1% vs. 35.1% · 95% CI [−22.8, −7.1]

H3 · CONTENT SHARING

+5.2 pp, n.s.

37.4% vs. 32.1% · 95% CI [−3.9, 14.4]

HOW TO READ

Each pair compares startups in the same industry classification; the gap is the social platform penalty for that service type. For other social platforms the gap was −3.9 pp (Model 3) and −5.1 pp (Model 4).

Table 2, Models 3–4 · social platform × personal record −0.640*; × content sharing 0.544* · * p < 0.05

4.1 MAIN RESULTS · TABLE 2

Results of Hypothesis Tests (H1–H3)

HOW TO READ

Each column is a logistic regression of Funded (0/1). Cells are coefficients, standard errors in parentheses; stars mark significance. Model 1: controls only. Model 2: + social platform (H1). Models 3 and 4: each adds one interaction (H2, H3).

HIGHLIGHTED ROWS

Social platform −0.2481*** (Model 2)
Social platform × Personal record −0.6402* (Model 3)
Social platform × Content sharing 0.5436* (Model 4)

The main effects of personal record and content sharing services are included in all models, so the interactions are not confounded with the effects of the service types themselves.

Table 2. Results of Hypothesis Tests (H1–H3)

Variable	Model 1	Model 2	Model 3	Model 4
Nonprofit (0/1)	−1.1555***	−1.1562***	−1.1538***	−1.1568***
	(0.1425)	(0.1425)	(0.1425)	(0.1425)
Pledge 1%	0.5368**	0.5319**	0.5310**	0.5322**
	(0.1693)	(0.1693)	(0.1693)	(0.1693)
# Industry categories	0.1823***	0.1888***	0.1889***	0.1888***
	(0.0086)	(0.0087)	(0.0087)	(0.0087)
ln # patents	0.8017***	0.8001***	0.8002***	0.8000***
	(0.0308)	(0.0308)	(0.0308)	(0.0308)
Female founder	0.5658***	0.5679***	0.5687***	0.5686***
	(0.0311)	(0.0311)	(0.0311)	(0.0311)
White founder	−0.8992***	−0.9021***	−0.9033***	−0.9046***
	(0.0655)	(0.0656)	(0.0656)	(0.0656)
Silicon Valley	0.1714***	0.1717***	0.1717***	0.1715***
	(0.0401)	(0.0401)	(0.0401)	(0.0401)
New England	0.2457***	0.2430***	0.2437***	0.2429***
	(0.0419)	(0.0419)	(0.0419)	(0.0419)
Personal record (Wellness/Health)	0.2580***	0.2429***	0.2499***	0.2418***
	(0.0339)	(0.0341)	(0.0343)	(0.0341)
Content sharing (Media & Ent.)	0.1309	0.1307	0.1284	0.0669
	(0.0802)	(0.0803)	(0.0803)	(0.0855)
Social platform (0/1)		−0.2481***	−0.2180***	−0.2818***
		(0.0596)	(0.0607)	(0.0617)
Social platform × Personal record			−0.6402*	
			(0.2841)	
Social platform × Content sharing				0.5436*
				(0.2352)
Constant	−0.7597***	−0.7502***	−0.7513***	−0.7460***
	(0.0714)	(0.0714)	(0.0714)	(0.0715)
Industry-group FE	Yes	Yes	Yes	Yes
Founding-year FE	Yes	Yes	Yes	Yes
Observations	51,738	51,738	51,738	51,738
Log likelihood	−28,960	−28,951	−28,948	−28,948

Note: Logistic regression. Standard errors in parentheses. All models include 43 industry-group fixed effects (excluding the four groups that overlap with the variable definitions) and founding-year fixed effects. N = 51,738. $^{+}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

43 industry-group FE · founding-year FE · N = 51,738

4.1 MAIN RESULTS · H4 SUPPORTED

Results of the H4 Test (Social Platform Startups)

33.4%
states exposure control

25.2%
does not

HOW TO READ

Predicted probability of raising funding (Model 6), comparing social platform startups only: those whose descriptions state that users can decide how much, and to whom, they are visible vs. those that do not. Gap ≈ 8.2 pp; odds about 55% higher.

Model 6	**β = 0.438, p = 0.042**
Model 7 (+ description length)	**β = 0.367, p < 0.10**

Table 3. Results of the H4 Test (Social Platform Startups)

Variable	Model 5	Model 6	Model 7
Nonprofit (0/1)	0.1178	0.0955	0.0708
	(0.6097)	(0.6131)	(0.6134)
Pledge 1%	−0.6859	−0.6291	−0.7198
	(1.1798)	(1.1797)	(1.1744)
# Industry categories	0.0931*	0.0874*	0.0797+
	(0.0414)	(0.0416)	(0.0417)
ln # patents	0.6981**	0.7197**	0.7351**
	(0.2455)	(0.2472)	(0.2479)
Female founder	0.3090*	0.2972+	0.2914+
	(0.1530)	(0.1533)	(0.1535)
White founder	−1.2099***	−1.2199***	−1.2183***
	(0.2807)	(0.2814)	(0.2815)
Silicon Valley	0.0847	0.0706	0.0707
	(0.2227)	(0.2235)	(0.2235)
New England	−0.3535	−0.3661	−0.3292
	(0.3521)	(0.3521)	(0.3526)
Personal record (Wellness/Health)	−0.2529	−0.2729	−0.3082
	(0.3132)	(0.3131)	(0.3142)
Content sharing (Media & Ent.)	0.6825**	0.6963**	0.7031**
	(0.2329)	(0.2333)	(0.2336)
User exposure control (0/1)		0.4378*	0.3671+
		(0.2156)	(0.2189)
Description length (1,000 chars)			0.2578*
			(0.1244)
Constant	−0.2819	−0.2973	−0.3846
	(0.3200)	(0.3206)	(0.3236)
Industry-group FE	Yes	Yes	Yes
Founding-year FE	Yes	Yes	Yes
Observations	1,804	1,804	1,804
Log likelihood	−961	−959	−957

Note: Logistic regression. Standard errors in parentheses. User exposure control follows the main definition (121 firms). Firms in founding-year or industry-group dummies that perfectly predict the outcome are dropped from estimation. N = 1,804. $^{+}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table 3 · social platform startups, N = 1,804

4.2 ROBUSTNESS CHECKS

Same direction across samples, definitions and measures

Check	Result
Linear probability model (Table A5)	Social platform −4.9 pp ($p < 0.001$); × personal record −0.127*; × content sharing 0.110*
Full sample incl. 2018 (Table A2)	H1 and H2 held; H3 and H4 in the same direction
Alternative definitions (Table A4)	Excluding Dating, or Social Network and Social Media only: H1–H3 held; H4 in the same direction
Description-based types (Table A6)	Showing content to others: interaction 0.249*; self-recording / sensitive information: −0.235, n.s., as predicted
Funding amount (funded firms)	Social platform startups raised about 57% less ($\beta = -0.856$, $p < 0.001$; N = 15,883)
User exposure control	LPM 0.083 ($p < 0.10$); broader definition (124 firms); dropping private community or visibility expressions one at a time

+ $p < 0.10$, * $p < 0.05$

Alternative explanations: kill zones and privacy regulation

Kill zone

Kamepalli, Rajan, & Zingales (2020)

The penalty should vary with the likelihood of competing with large platforms, unrelated to information sensitivity. Yet it was greatest for personal record services (H2), content sharing services were indistinguishable (H3), and exposure control features, unlikely to reduce competition risk, helped (H4).

GDPR (May 2018)

Jia et al. (2021); Janßen et al. (2022)

Effects appeared mainly among European ventures; our sample consists of U.S. startups. Regulatory costs common to an industry group are absorbed by the fixed effects, and the account alone has difficulty explaining why the penalty shrank for content sharing services (H3).

A response to the incident?

Section 4.3.1

Adoption of exposure control features was 6.7% among firms founded before the incident and 6.4% after, and the effect did not differ (interaction $\beta = -0.249$, $p = 0.660$). The features did not become more common after the incident.

The influence of market structure cannot be ruled out entirely (Section 4.3.2)

V

Discussion and Conclusion

5.1 Discussion

5.2.1 Theoretical Contributions

5.2.2 Practical Implications

5.2.3 Limitations and Future Research

5.1 DISCUSSION

H2 and H3 concern the size of the harm should the same problem occur; H4 concerns the likelihood that it occurs at that firm

H1	Category membership can be both a foundation of legitimacy and a channel of cost for startups: as long as investors use categories as frames for understanding firms, suspicion attached to those frames also enters their evaluations.
H2 · H3	Evaluators' suspicion does not attach uniformly to the category label but is adjusted according to how much harm the risk revealed by the incident would cause if it recurred in a firm's service.
H4	Attracting less stigma does not necessarily require a response tailored to the incident; service features visible to investors can by themselves make a difference.

5.2.1 THEORETICAL CONTRIBUTIONS

Three contributions

1 **Category spillover research**

We extend the focus of category spillover research from the losses of established firms to the **resource acquisition of new ventures**.

2 **Who suffers more from a spillover**

The criterion of deductive generalization (Naumovska & Zajac, 2022) can shift from **similarity to the offender** toward **relevance to the harm the incident revealed**, connecting contextual integrity (Nissenbaum, 2004) with category spillover research.

3 **Stigma transfer research**

Startups that lack collective means and cannot leave their category still attract less stigma when their **self-presentation shows that they lack the flaw** behind the incident.

A category penalty is not fixed

Founders

For services that hold users' sensitive information, building features that let users set their own exposure and communicating them clearly can be a way to stand apart within the category.

Investors

Uniformly discounting an entire category can mean missed opportunities.

Policy

Certification or disclosure schemes that make design-level safeguards externally verifiable could help limit the damage that spreads to blameless startups.

Limitations and future research

- Whether a funding round was recorded → funding amounts, stages, and timing
- One event, one category, U.S. startups → other categories, e.g., healthcare diagnostics or crypto-asset startups
- Descriptions observed at data collection → archived websites, app store descriptions, privacy policies
- Investors' interpretations inferred from outcomes → interviews or experiments with investors

A category is a starting condition that startups cannot choose, *but evaluations within the same category differ according to what investors see in each firm.*

Thank you.

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APPENDIX

Description-Based Validation of Service Types

Table A6. Description-Based Validation of Service Types

Panel A. Share of firms whose descriptions contain each set of expressions, by social platform type

Social platform type	N	Self-recording / sensitive information	Shared content
Personal record services (Wellness/Health)	71	28.2%	19.7%
Content sharing services (Media & Ent.)	101	4.0%	67.3%
Other social platforms	1,648	2.6%	31.1%

Panel B. Moderating effects of service types measured from description expressions

Description-based service type	Social platforms with expressions	Main effect	Social platform × service type
Self-recording / sensitive information	67	0.418***	−0.235
		(0.042)	(0.281)
Shared content	594	0.328***	0.249*
		(0.035)	(0.121)

Note: Panel B is a logistic regression (N = 51,738) including the control variables (including the personal record and content sharing classifications), industry-group fixed effects, and founding-year fixed effects. Standard errors in parentheses. Self-recording / sensitive-information expressions: track your/their/my, journal, diary, mood, symptom, habit, personal health, health record, medical record, self-care, sleep, mental health, therapy, therapist, support group, peer support, coping, grief, addiction, anxiety, depression, LGBT, sexual, illness, disease, patient, cancer, diabet-, pregnan-, fertility. Shared-content expressions: stream, video, music, creator, follower, fans, audience, broadcast, influencer, entertain-, podcast, watch. $^{+}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

APPENDIX

Alternative definitions and linear probability models

Table A4. Results under Alternative Definitions of Social Platforms

Definition of social platform	H1	H2	H3	H4
Main definition	−0.2481***	−0.6402*	0.5436*	0.4378*
	(0.0596)	(0.2841)	(0.2352)	(0.2156)
Excluding Dating	−0.2619***	−0.6386*	0.5589*	0.4674*
	(0.0605)	(0.2913)	(0.2355)	(0.2188)
Social Network and Social Media only	−0.2754***	−0.6492*	0.5623*	0.4239+
	(0.0612)	(0.2986)	(0.2379)	(0.2537)

Note: Logistic regression, including the control variables, industry-group fixed effects, and founding-year fixed effects. Standard errors in parentheses. H1 is the main effect of social platform status; H2–H3 are interaction terms; H4 is estimated within social platforms under each definition. $^{+}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table A5. Comparison with and without Industry-Group Fixed Effects (Linear Probability Model)

Variable	(1)	(2)	(3)	(4)
Social platform	−0.075***	−0.049***	−0.044***	−0.055***
	(0.010)	(0.011)	(0.011)	(0.011)
Social platform × Personal record			−0.127*	
			(0.053)	
Social platform × Content sharing				0.110*
				(0.051)
Constant	0.348***	0.349***	0.348***	0.349***
	(0.016)	(0.016)	(0.016)	(0.016)
Controls	Yes	Yes	Yes	Yes
Industry-group FE	No	Yes	Yes	Yes
Founding-year FE	Yes	Yes	Yes	Yes
Observations	51,738	51,738	51,738	51,738
R-squared	0.077	0.113	0.113	0.113

Note: Linear probability model. Robust standard errors in parentheses. N = 51,738. $^{+}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

APPENDIX

Exposure control expressions and the full sample

Table A3. Expressions Used to Measure User Exposure Control

Design type	Meaning	Expressions
Anonymous or pseudonymous participation	Users can participate without revealing their real names	anonymous, anonymity, anonymously, pseudonym
Invitation-based or private community	Spaces visible only to designated members	invite-only, invite only, invitation-only, invitation only, by invitation, members-only, members only, private communit-, private network, private social, private group, private space, closed communit-, closed network, closed group, safe space, safe place, safe environment, safe community, safe platform
Visibility settings	Users decide what to show and to whom	privacy control, privacy setting, control who, control what, control over, choose who, choose what
Industry classification	Private service by definition	Private Social Networking classification

Note: User exposure control is a design that lets users decide how much, and to whom, they are visible. The list of expressions was defined by design type before any results were examined, and it was fixed after reading the actual descriptions and removing expressions captured in unrelated contexts. All expressions were searched in lowercased descriptions. Firms by type: anonymous or pseudonymous participation 29, invitation-based or private community 26, visibility settings 11, Private Social Networking classification 63. Six firms fall into two or more types, and 121 firms fall into at least one.

Table A2. Robustness Check: Full Sample Including Firms Founded in 2018 (N = 61,618)

Hypothesis	β	*p*
H1 Social platform	−0.256	< 0.001
H2 Social platform × Personal record	−0.692	0.007
H3 Social platform × Content sharing	0.328	0.142
H4 User exposure control (N = 2,104)	0.324	0.104

Note: Logistic regression, including the control variables, industry-group fixed effects, and founding-year fixed effects.

ANTICIPATED QUESTIONS · DESIGN

Identification and sample

Q1. How do you know the penalty comes from the scandal, not from social platforms in general?

Social platforms are compared with other startups in the same industry groups, so structural differences across groups are absorbed by fixed effects. The penalty also varied with the harm the revealed problem would cause (H2, H3) and with exposure control (H4), a pattern that kill-zone and regulatory accounts alone have difficulty explaining; the influence of market structure cannot be ruled out entirely.

Q2. Why exclude firms founded in 2018?

About 44% of firms record only a founding year, with the date entered as January 1, so for 2018 founders it is impossible to tell whether they were founded before or after March 2018. In the full sample including them (N = 61,618), H1 and H2 held and H3 and H4 were in the same direction (Table A2).

Q3. Recently founded firms had less time to raise funding.

Founding-year fixed effects absorb year-to-year differences in the investment market as well as right-censoring, whereby more recently founded firms had less time to raise funding before data collection.

ANTICIPATED QUESTIONS · MEASUREMENT

Service types and exposure control

Q4. Are self-registered industry classifications valid measures of service type?

A registered classification is a self-declared marker of the service a firm provides. Descriptions confirm the split: self-recording or sensitive-information expressions appear in 28.2% of personal record platforms vs. 4.0% of content sharing; shared-content expressions in 67.3% vs. 19.7%. Description-based measures yield results in the same direction (Table A6).

Q5. How was user exposure control measured? Could the list be cherry-picked?

Expression lists for three design types were defined before seeing any results; we read the captured descriptions, removed expressions captured in unrelated contexts, fixed the list, and generated the variable only once. The measure was cross-validated against each firm's website (Table A3).

Q6. Descriptions were observed at data collection, not at founding.

This is a limitation; archived websites, app store descriptions, or privacy policies could capture product features at founding more directly. Adoption did not rise after the incident (6.7% vs. 6.4%), and the effect did not differ between the two groups ($p = 0.660$).

ANTICIPATED QUESTIONS · RESULTS

Small groups and the strength of the effects

Q7. Personal record and content sharing platforms are small groups (71 and 101).

We report confidence intervals for the predicted differences (personal record: −22.8 to −7.1 pp). The interactions were significant in the linear probability model and under alternative definitions of social platforms (Tables A4, A5). In the full sample, H3 was in the same direction but not significant ($p = 0.142$).

Q8. Perhaps health-related startups simply raise less?

The comparison is with other startups in the same classification, and the classification itself was favorable for fundraising (other startups: 35.1%). The penalty can be interpreted as arising not from the field but from the combination of suspicion toward the category and a service that relies on sensitive information users entrust for their own use.

Q9. How robust is the H4 result?

$\beta = 0.438$ ($p = 0.042$); with description length controlled, $\beta = 0.367$ ($p < 0.10$). It was positive in the LPM (0.083, $p < 0.10$), under a broader definition (124 firms), and when private community or visibility setting expressions were dropped one at a time; in the full sample, in the same direction ($p = 0.104$).

ANTICIPATED QUESTIONS · ALTERNATIVES AND SCOPE

Competing explanations and generalizability

Q10. Isn’t this a kill-zone effect?

Under the kill-zone account (Kamepalli et al., 2020), the penalty should vary with the likelihood of competing with large platforms and be unrelated to information sensitivity. Yet content sharing services, which overlap more with Facebook’s core business, were indistinguishable from other startups, and exposure control features, unlikely to reduce competition risk, were associated with funding.

Q11. Could GDPR explain the result?

GDPR effects appeared mainly among European ventures, and Jia et al. (2021) used U.S. ventures as the comparison group. Our sample consists of U.S. startups, regulatory costs common to an industry group are largely absorbed by the fixed effects, and regulation alone has difficulty explaining why the penalty shrank for content sharing services.

Q12. Do the findings generalize beyond this case?

The context serves as a case for testing a category-level theory. We studied one event, one category, and U.S. startups; testing the same question in other categories that have experienced an incident at a leading firm, such as healthcare diagnostics or crypto-asset startups, would establish the scope of the findings.